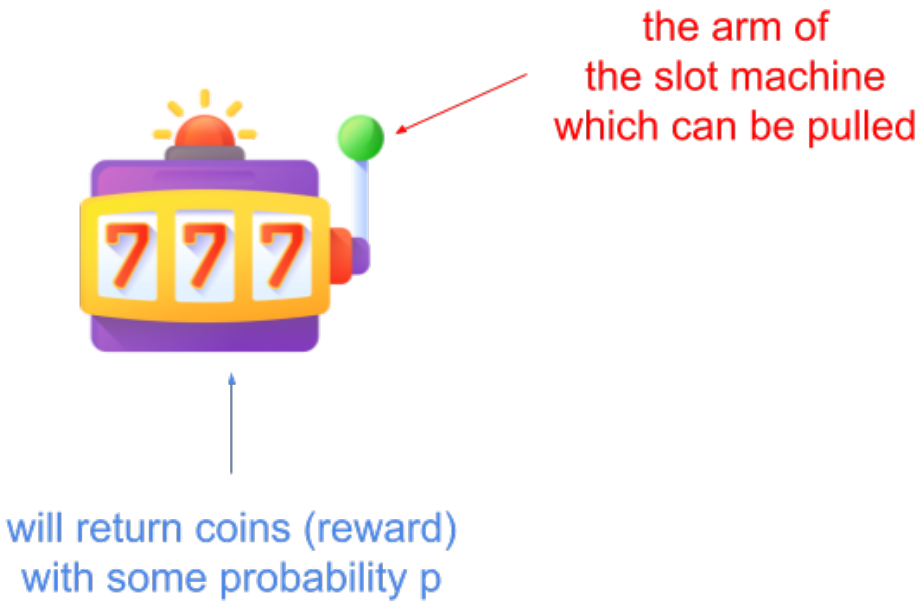


Figure 5.1 A slot machine with an arm that can be pulled. The machine may return coins according to its reward probability when its arm is pulled.

Upon pulling the arm of the slot machine, you may receive coins as reward; however, there's randomness in this process. Specifically, programmed within the inner-workings of this slot machine is a reward probability p. Each time the arm of the machine is pulled, the machine returns coins according to that probability. If $p = 0.5$, then the machine rewards its players half of the time. If $p = 0.01$, then roughly only one in one hundred pulls will result in coins being returned. Since this probability is programmed inside the machine and therefore hidden away from us, there's no way for us to know for sure what that probability is.

In this hypothetical setting, the casino programs its slot machines so that the rate at which these machines reward players with coins is lower than that of the machines being played and receiving coins. In other words, even if there are occasional winners who receive rewards from these slot machines, on average, it's the casino that makes a profit.

IMPORTANT **Why is it called a bandit?**

A particularly unsuccessful player who is losing coins faster than winning them might begrudgingly call the slot machine he's been playing with "a bandit," thinking that the machine is stealing his money. Since the machine has an arm that may be pulled, it can be called a one-armed bandit.

Now, imagine that there's not just one single slot machine, but a *row* of machines that we may choose to play with, each with its own reward probability p, as visualized in Figure 5.2.

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